MILITARY SPECIFICATION

WELDING, SPOT, INERT-GAS SHIELDED ARC

1. SCOPE

1.1 This specification covers requirements for the tungsten inert gas shielded arc spot welding process.

1.2 Classification. Classification shall be based on the function of the arc spot welded joint. Welds shall be classified as follows:

Class 1 - Used in joints the failure of which would reduce the overall strength of the weapons system or preclude the intended functioning or use of equipment.

Class 2 - A weld which is considered noncritical and for which no stress analysis is requested.

1.2.1 The class of welding shall be designated on the item specification or drawing.

1.2.2 The shear strength values for both class 1 and class 2 shall be in accordance with table I unless alternate values are established (see 1.2.2.1) and are specified on the drawing or specification for the item.

1.2.2.1 Alternate sheer strength values. Minimum and average strengths higher than those shown in table I may be established provided the higher strengths are maintained in all tests specified herein.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitations for bids form a part of this specification to the extent specified herein.

DRAWINGS

Military

MS33528	Thickness, Flat Rolled Metal Products, Less Than One-Fourth Inch

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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SHEAR STRENGTH
FOR ARC SPOT WELDS

TABLE I

Thinner Sheet Thickness / Thickness per MS33528	Minimum Shear Strength per Spot (lb)	Minimum Average Shear Strength (lb)
0.006	54	67
0.008	90	112
0.010	123	154
0.012	165	206
0.016	240	300
0.018	294	367
0.020	318	397
0.022	366	457
0.028	513	641
0.032	627	784
0.036	753	941
0.040	876	1095
0.045	1077	1346
0.050	1275	1594
0.056	1530	1912
0.063	1854	2317
0.071	2238	2798
0.080	2646	3308
0.090	3054	3817
0.100	3432	4290
0.112	3819	4774
0.125	4248	5310

3. QUALITY STANDARDS

3.1 The supplier (plant at which the equipment is located) is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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3.2 Appearance. The outer surfaces of all spot welds shall be smooth and free of cracks. The concavity on each of the outer surfaces shall not exceed 0.005 inch or 20 percent of the thickness of the sheet in which the concavity occurs, whichever is less. The convexity on each of the outer surfaces shall be controlled as required for the function of the part.

3.3 Penetration and internal defects. Weld penetration and soundness shall be determined by examination of a metallurgical section through the center of the welds in test specimens welded according to the production welding schedule. The outline of the fused zone shall be generally consistent in size and shape.

3.3.1 Internal defects. The penetration into the bottom sheet shall be equal to or greater than 20 percent of its thickness or 40 percent of the top sheet thickness, whichever is less, over an area whose major axis is 80 percent of the weld diameter at the interface. The total area of voids, blowholes and porosity shall not exceed, for class 1 welds 15 percent and for class 2 welds 25 percent of the area of the weld nugget, nor extend within 15 percent (of the weld diameter at the faying surface) of the boundaries of the fused nugget structure (see figure 1).

3.3.2 Sheet separation. Sheet separation at the edge of the weld shall not exceed 10 percent of the average thickness of the two members plus the original gap (not to exceed 0.010 inch, see 5.3). Sheet separation shall be measured at a distance from the edge of the weld approximately equal to one half the weld diameter.

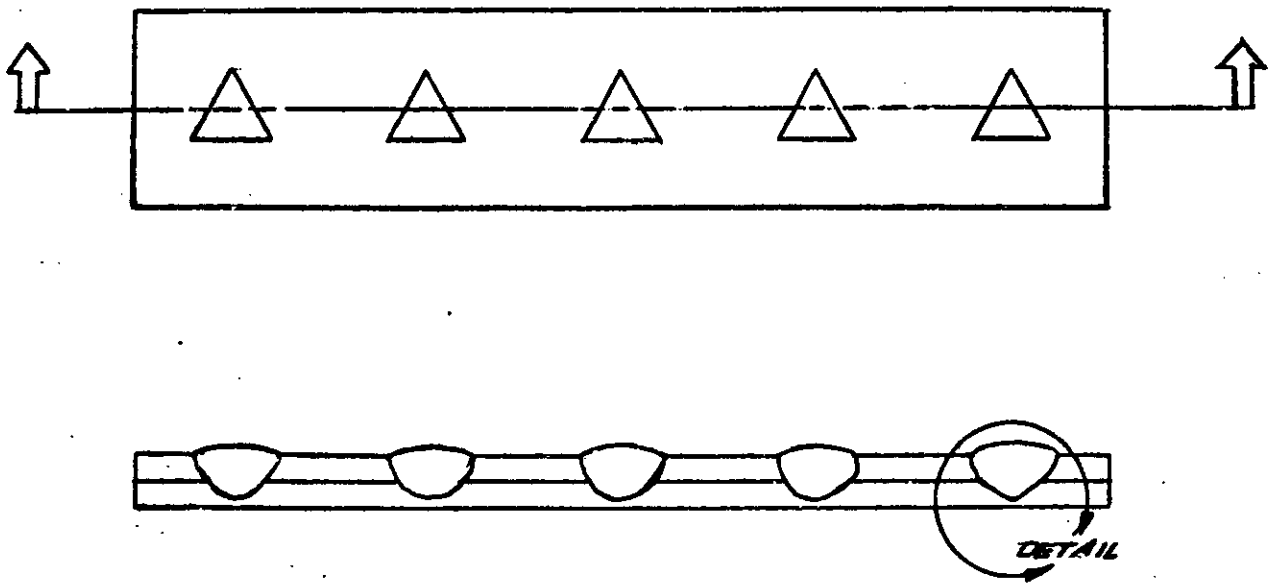
3.4 Shear strength. The ultimate shear strength of every spot weld specimen shall be equal to or exceed the applicable minimum values in table I or the minimum values established under 1.2.2.1.

3.4.1 Unequal thickness combination. In a joint between sections of unequal thickness, or of alloys of different strengths, the minimum shear strength requirements shall be based on the thickness of the thinner sheet or the strength of the weaker material, whichever requirement is lower.

4. EQUIPMENT

4.1 The welding machine shall consist of a suitable source of electrical energy, an adequate electrode and holder and a means of reliably controlling and indicating, directly or indirectly, the magnitude of the current and the time of current flow. Inert gas shielding shall be provided for controlled time periods before and after current flow. When filler material is used, the initiation, rate and time of material feed shall be included as machine controls. The degree of accuracy of all controls shall be as required to produce welds meeting requirements of this specification.

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(1) FOR CLASS "1" .15 D; FOR CLASS "2 .25 D
NO DEFECT PERMITTED IN SHADED AREA.
d .8 D

METALLURGICAL SPECIMEN
FIGURE 1

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4.1.1 Control settings. The equipment shall be provided with suitable means for indicating directly or indirectly the settings for each variable which must be controlled in order to make satisfactory welds. The control equipment shall operate consistently so that the welds will meet the requirements specified herein for the qualification tests.

4.2 Shear testing machines. The contractor shall provide spot weld shear testing machines as required. All shear testing machines shall be accurate within ±2 percent of the indicated reading. Portable spot weld shear test machines shall be checked for accuracy at intervals not to exceed 2 months.

5. MATERIALS

5.1 Permissible alloys. Alloys welded in accordance with this specification shall include the low carbon (D max = 0.2 percent) steels, austenitic stainless steels, precipitation hardening corrosion resistant steels, cobalt alloys, nickel base heat resisting alloys, titanium and titanium alloys. All metals and combinations of metals shall conform to the requirements of 5.1.1. Welding of other materials shall be considered outside the scope of this specification and shall be in accordance with the contract or order.

5.1.1 All metals welded in production shall require establishment of weld ductility characteristics. Such welded material shall exhibit a tension strength not less than 25 percent of the minimum shear strength requirements of table I. Under conditions where the arc spot welded production part will be subjected to subsequent thermal treatment the welds shall exhibit minimum tension strength, as required herein, before (as welded) and after thermal treatments. The tension (tensile pullout) configuration may be of any type which will satisfactorily apply tension load to the arc spot weld. Evaluation may be conducted on one material thickness only. Materials which satisfactorily meet this requirement may be welded in production without further tension strength evaluations. If tension strength levels established herein are not obtained, special thermal treatments may be applied provided such treatments produce the required tension strength levels.

5.2 Surface condition. The surfaces of the parts to be welded shall be free from objectionable films such as oxides, scale, ink, grease, dust or other substances detrimental to the welding process.

5.3 Mating parts. Mating parts, assembled for welding, shall fit so that before the first and each successive weld is made the surfaces to be joined are in contact. Where engineering considerations permit, on combinations having each outer member 0.030 inch or thicker, faying surface gaps up to 0.010 inch maximum will be permitted, provided the gap condition is duplicated in the specimens used to establish the welding schedules.

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6. QUALIFICATION OF WELDING MACHINES

6.1 Welding machines. Qualification tests of welding machines shall be performed to determine the consistency of the machines at or near the desired operating range. The machine qualification data shall furnish the information specified in figure 2.

6.1.1 Test materials. The alloy most frequently welded in production shall be used in the qualification tests. The selection of the alloy is at the option of the contractor.

6.2 One machine of each distinctive type shall satisfactorily pass the complete qualification tests before any part or assembly welded on the machine is considered acceptable as an item on a contract.

6.2.1 Types of equipment. Distinctive types of equipment are those differing in any of the following respects:

a. Manufacturer of machine or control panel, or type of machine or model number.

b. Electrical rating or capacity.

c. Type of power supply.

6.2.1.1 Additional machines of the same distinctive types as an approved machine may be qualified by the establishment of satisfactory production settings for each machine.

6.2.2 Requalification. When the equipment has once been qualified, it need not be requalified for future contracts or production provided no change in basic material or range of settings is involved. A change of location within the plant not involving a change in power line, does not necessitate requalification. Requalification shall be required if the machine is rebuilt or if significant operational changes are made therein. Approval of machines qualified under superseded specifications shall be honored except as specifically provided herein.

6.3 Test specimens. A minimum of two sets of test specimens will be required, one at each limit of the range of machine settings for which qualification is desired. Except when the option of 6.3.1 is used, each set shall consist of not less than 105 consecutive welds consisting of single spot specimens as specified in figure 3 or multiple spot specimens as specified in figure 4. No maintenance attention or control adjustments are permitted during the welding of a set of specimens.

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Company _____ Date _____

Manufacturer of Equipment: Machine _____ Control _____

MATERIAL & VARIABLES:

MATERIALS:

Type: _____ H.T. Cond. _____, Cleaning _____,
Thickness; _____, to _____,
Filler Material; _____, Size _____, Shape _____.
Other:

WELDING VARIABLES:
Tungsten; Type _____, Size, _____, Shape _____
Gas; Type _____, Flow Rate _____
Preflow Time _____ Postflow Time _____
Welding Current:
Polarity; _____, Initial _____ Amps, Weld _____Amps,
Final _____Amps, Preheat _____Amps, Postheat _____
Welding Time; Initial Slope Time _____, Weld Time _____,
Postheat Time _____, Final Slope Time _____,
Tailing Time _____, Other _____.
Filler Material; Wire Start Delay _____, Feed Time _____,
Start; Hot _____ Soft _____
Nozzle Size; _____
Joint Preparation: _____

SIGNATURES: Operator _____ Inspector _____

TEST RESULTS:

SHEAR TEST:

1	16	31	46	61	76	91
2	17	32	47	62	77	92
3	18	33	48	63	78	93
4	19	34	49	64	79	94
5	20	35	50	65	80	95
6	21	36	51	66	81	96
7	22	37	52	67	82	97
8	23	38	53	68	83	98
9	24	39	54	69	84	99
10	25	40	55	70	85	100
11	26	41	56	71	86	
12	27	42	57	72	87	
13	28	43	58	73	88	
14	29	44	59	74	89	
15	30	45	60	75	90	

Average Shear Strength _____
Minimum Shear Strength _____
Required Average (Table I) _____
Required Minimum (Table I) _____
Variation _____

METALLOGRAPHIC:

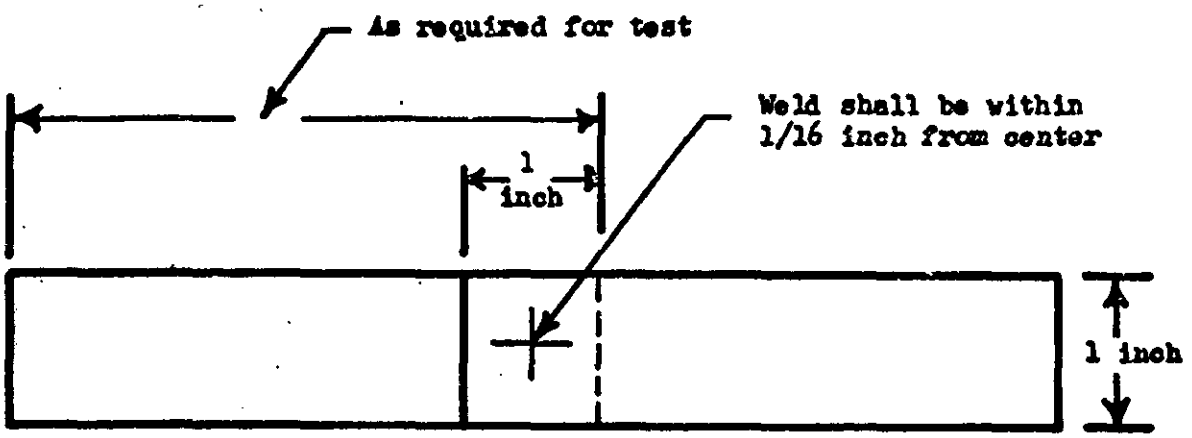
1 _____
2 _____
3 _____
4 _____
5 _____

Average Nugget Diam. _____

SIGNATURES:
Welding Supervisor _____
Test Lab Representative _____
_____

FIGURE 2 MACHINE QUALIFICATION DATA

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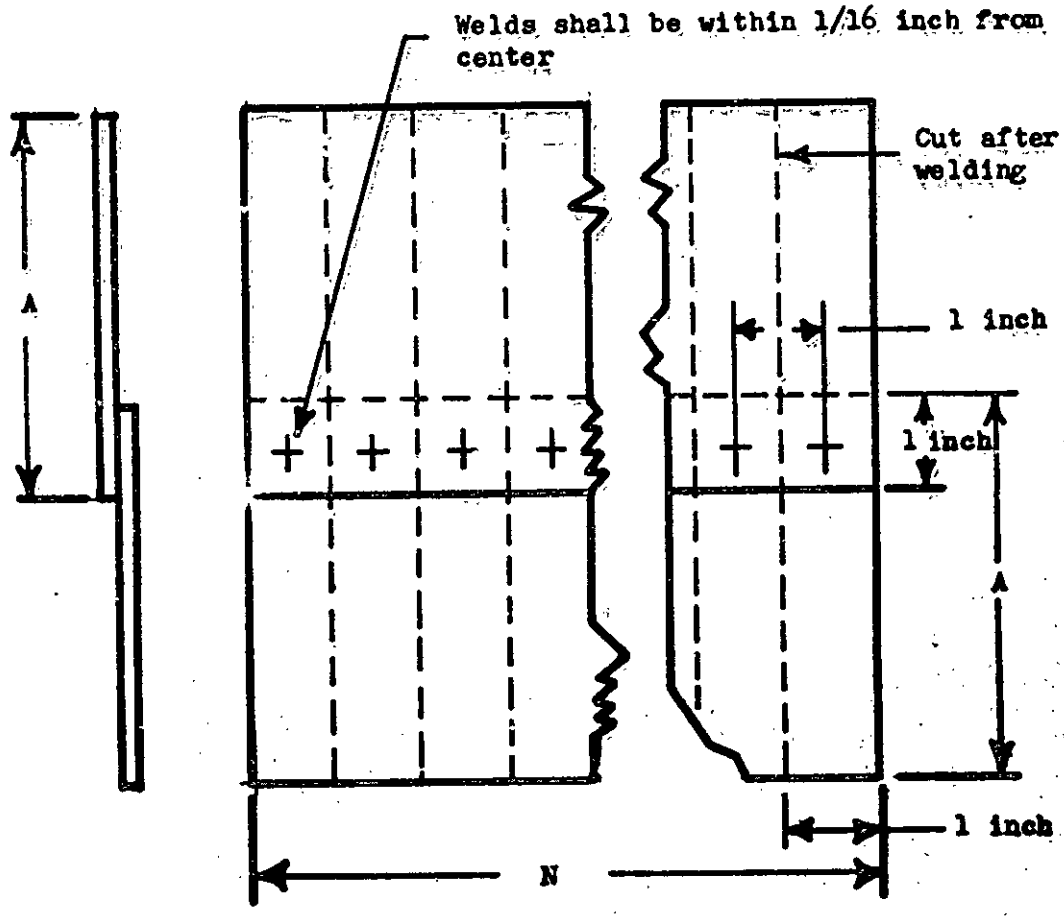


The above dimensions may be increased when sheet thickness is above 0.100 inch and when the higher strength values are used in accordance with 1.2.2.1.

Figure 3

Single spot shear specimen

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N = Number of welds.
Dimension "A" shall be as required by test.
The above dimensions may be increased when sheet thickness is above 0.100 inch and when the higher strength values are used in accordance with 1.2.2.1.
Panel shall be cut into single spot specimens after welding.

Figure 4

Multiple spot shear specimen

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6.3.1 Optional qualification procedure. When equipment identical to the equipment being qualified has been previously tested by the manufacturer of equipment and the results obtained indicate compliance with the requirements specified herein, the number of specimens required for each set in 6.3 may be reduced to 30, 25 for tensile test and 5 for macroscopic examination. The report on the tests made by the equipment manufacturer shall be made a part of the qualification tests specified herein, when this option is used.

6.4 Tests

6.4.1 Preparation of specimens. After welding, all multiple spot specimens shall be cut into single spot test coupons conforming to figure 3 prior to testing.

6.4.2 Tests of specimens. All single spot specimens except those required for metallurgical examination shall be loaded to determine the shear strength of the weld.

6.4.3 Metallurgical examination. Five single spot specimens, chosen from each set, shall be cross-sectioned as closely as possible through the center of the weld, and then polished and etched for metallurgical examination.

6.4.4 Shear strength requirements. The shear strength of each weld shall be in accordance with the minimum requirements of table I or the higher minimum values established under 1.2.2.1. The failure of one spot weld to meet the minimum strength shall be cause for rejection of the entire set, and the welding methods or equipment shall be modified and the test repeated for the particular combination involved.

6.4.4.1 The average strength of each set of welds shall equal or exceed the lowest acceptable average strength value shown in table I or the higher minimum average values established under 1.2.2.1.

6.4.4.2 A maximum variation of 0.25 (calculated using sample range as outlined in section 9) shall be permitted in the shear values for machine qualification.

6.4.5 Metallurgical requirements. Weld penetration, fusion and internal or external defects shall conform to the applicable requirements of section 3.

6.5 Machine and process data. The suggested form, figure 2 or equivalent, shall be completed in duplicate, one copy to be posted on or near the machine. The original is to be filed and maintained as a permanent record of machine operation.

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7. ESTABLISHMENT OF SPOT WELDING PROCESS OR SCHEDULE

7.1 Welding schedule. Suitable welding schedules shall be established for each alloy or combination of dissimilar alloys and each gage combination to be spot welded in production with each machine under consideration. Gage combinations falling within both of the following limits shall not require separate welding schedules:

a. Twenty percent variation in thickness of any outer sheet.

b. Ten percent in the overall thickness of the combination to be welded.

7.1.1 The welding schedule shall be established prior to the welding of production parts and shall include the procedure for preparation of materials (including cleaning practices), and all details of machine setup and control settings for each machine to be used in production welding. The suitability of the schedule shall be demonstrated by making and testing of not less than the number of consecutive welds required by table II. The welded specimens shall conform to figures 3 and 4.

TEST SPECIMENS FOR ESTABLISHING WELDING SCHEDULES

TABLE II

	Number of Shear Strength Specimens	Number of Metallurgical Specimens	Number of other specimens (see para 7.2.3.1)
Class 1	10	3 (Macro-6-12x Mag)	6 Nugget Diam.
Class 2	5	2 (Macro-6-12x Mag)	4 Nugget Diam.

7.2 Tests

7.2.1 Shear specimens. Preparation and tests of specimens shall conform to the applicable requirements in section 6. When multiple thicknesses are welded, the load in the shear tests shall be applied to the two outer sheets of the joints, unless otherwise stated on the drawing for the part. The required number of welds, as specified in table II, shall be failed in shear.

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7.2.2 Shear strength requirements. The minimum and average shear strengths shall meet the requirements of table I or the requirements established under 1.2.2.1. The failure of one weld to meet the minimum requirements shall be cause for rejection of the entire set, and the welding methods or equipment shall be modified and the test repeated for the particular combination involved.

7.2.2.1 The variation in shear strength shall not exceed 0.25 for each group of shear strengths and shall be calculated in using the sample range as outlined in section 9.

7.2.3 Metallurgical requirements. The metallurgical specimens, as specified in table II, shall be examined for weld penetration, fusion and internal or external defects conforming to the applicable requirements of section 3.

7.2.3.1 Nugget diameters. When the use of nugget diameters is planned for production quality control, the diameter of each weld, as specified in table II, shall be measured at each faying plane and recorded. The records shall show the average and minimum diameters. This will establish the average and minimum nugget diameters for production quality control.

7.3 Records. There shall be made readily available to the machine operator and the authorized inspection personnel, for reference at any time, records showing the welding machine settings. These settings shall be for all variable controls and minimum and average spot shear strengths or nugget diameters for all alloys and thickness combinations welded on that machine.

8. PRODUCTION QUALITY CONTROL

8.1 General. Production spot welding shall be so accomplished to obtain welds of uniform strength above the accepted minimum and of acceptable metallographic structure. All test specimens selected for checking production spot welding shall be representative of manufacturing practice. Specimens shall be tested under the surveillance of personnel authorized by the supplier. Prior to spot welding, all production parts shall be checked for conformance to the applicable requirements of section 5.

8.2 Schedules. Competent personnel appointed by the supplier shall be responsible for the control of machine settings and all welding schedules.

8.3 Maintenance of equipment. Each item of equipment shall be inspected periodically. Adequate maintenance shall be furnished, and defective parts affecting machine operation shall be replaced before production is resumed.

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8.4 Routine check specimens. Using the established setting, a 3-spot metallurgical specimen conforming to figure 1, or 3 shear coupons, or a simulated section of the production joint shall be welded and examined at the following times:

a. At the start of each welding schedule.

b. At intervals not to exceed two hours and at the end of a production run if more than an hour has elapsed since the last check.

c. Upon changing the size or configuration of the electrode.

d. At the request of Quality Control.

A record of these checks shall be maintained.

8.4.1 The test specimens shall conform to the production parts they represent with respect to material thickness combination, surface condition and machine settings. The heat sink characteristics of the test specimen weld fixture shall be, as nearly as possible, the same as the production weld fixture.

8.5 Control adjustment. When routine check tests indicate that adjustment of the control settings are desirable, the settings may be varied by ±5 percent from the established values or by ±10 percent when only one variable is adjusted. Verification of the adjusted schedule shall consist of a routine check test.

8.5.1 If satisfactory welding cannot be done within the limits of adjustment stated in 8.5, the machine shall be checked for faulty operation. If it can be determined that a machine malfunction was the cause of the faulty welds, and upon the correction of the malfunction satisfactory welds can be made, it will not be necessary to conduct tests to re-establish the settings.

8.6 Inspection of routine check specimens. All welds made for routine checks shall conform to the weld quality requirements of section 3.

8.6.1 The average and minimum shear strengths shall equal or exceed the appropriate values of table I or the minimum values established under 1.2.2.1. After testing in shear, each of the fractured welds shall be visually examined for fusion and evidence of obvious internal defects, such as cracks and porosity. The variation in shear strength shall conform to 7.2.2.1.

8.6.2 If a routine check indicates that the welds made during the preceding period are unsatisfactory, the welds made in that period shall be rejected and submitted for salvage action.

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8.7 Inspection of welded parts. All welds shall conform to the weld quality specified in section 3.

8.8 External defects. Production parts shall be visually examined for the presence of external defects specified in section 3. No external cracks shall be acceptable.

8.9 Internal defects. Inspection of production parts for internal defects will not normally be required. When deemed necessary by the authorized inspector, samples shall be tested of actual production work, in conformance with the contract or specification for the part.

8.10 Rejection and salvage. Any part containing arc spot welds with defects over and above the limits on permissible defects specified herein shall be rejected.

9. NOTES

9.1 Intended use. This specification covers the requirements for equipment, materials, methods of preparing the material and methods of weld process control in inert-gas shielded arc spot welding of corrosion and heat-resisting alloys.

9.2 Estimating variation.

$$\text{Variation} = \frac{\text{Range}}{\text{Average}} = \frac{\text{Difference between largest and smallest shear strength values}}{\text{Average of individual shear strengths}}$$

9.2.1 Sample calculation.

Shear strengths in pounds = 2200, 2170 and 1790

Total in pounds = 6160

Average in pounds $= \frac{6160}{3} = 2053$

Range in pounds = 2200 - 1790 = 410

Variation $= \frac{\text{Range}}{\text{Average}} = \frac{410}{2053} = 0.2001$

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